

stock, this indicator will enable market participants to trade in a more informed manner. The indicator will be included in SIP data that is disseminated by the exclusive SIPs and consolidated market data³²¹ disseminated by competing consolidators, which will help to ensure the wide availability of information about the applicable minimum pricing increment for each NMS stock.

One commenter supported the minimum pricing increment indicator stating that it would make the new increments easier to implement.³²² This commenter suggested that the exclusive SIPs publish the indicator every morning, in a machine-readable format, and free of charge.³²³

The exclusive SIPs currently provide certain information that comprises regulatory data as part of SIP data.³²⁴ Under the rule, the exclusive SIPs will be required to collect and disseminate a new regulatory data element, the minimum pricing increment indicator, and collect and disseminate this regulatory data element as part of SIP data.³²⁵ To the extent that the exclusive SIPs charge fees for this new regulatory data element, such fees will be required to be filed under rule 608 of Regulation NMS and must be fair and reasonable and not unreasonably discriminatory.³²⁶ The Commission has not required a specific format for SIP data, including regulatory data; such format will be developed by the Operating Committees' for the Equity Data Plans consistent with regulatory requirements.³²⁷

³²¹ See rule 600(b)(24).

³²² See MMI Letter at 6.

³²³ See MMI Letter at 6.

³²⁴ See MDI Adopting Release, supra note 10, at 18729 for a description of the regulatory messages that are disseminated by the exclusive SIPs.

³²⁵ See Rule 600(b)(89)(iv).

³²⁶ Sections 11A(c)(1)(C) and 11A(c)(1)(D) and Rule 603(a). See MDI Adopting Release, supra note 10, at 18684.

³²⁷ The SIP data format will be available at the SIP's web site, <https://www.ctaplan.com/index>.

D. Minimum Pricing Increment for Trades

The Commission proposed to amend Rule 612 to introduce minimum pricing increments for trades of NMS stocks where the minimum pricing increment for trading NMS stocks priced at or above \$1.00 would vary and correlate to one of the four proposed minimum pricing increments for quoting (i.e., \$0.001, \$0.002, \$0.005 and \$0.01), subject to proposed exceptions for midpoint trades and benchmark trades.³²⁸ The proposed minimum pricing increments for trades would have harmonized trading increments (1) with the proposed variable minimum pricing increments for quotes and orders (in this release, “Quote and Trade Harmonization”), and (2) across all trading venues (in this release, “Venue Harmonization”). Specifically, Quote and Trade Harmonization would have required all trading to occur in the same increments as those required of quotes and orders subject to certain exceptions for midpoint and benchmark trades. Venue Harmonization would have required all trading on exchanges, ATSS and OTC to occur in the same pricing increments.

In the Proposing Release,³²⁹ the Commission stated that it was concerned about the competitive dynamic between exchanges, ATSS, and OTC markets and that a potential contributing factor was the ability of OTC market makers to execute orders in price increments that are smaller than the price increments that exchanges and ATSS can practically provide.³³⁰ The Commission stated that applying the minimum pricing increment to trades across all venues would promote equal regulation and fair competition among market participants such as

³²⁸ The Commission also proposed to impose a minimum pricing increment for trades for quotes and orders priced less than \$1.00 that would have been the same as the minimum pricing increment for quotes, i.e., \$0.0001. See Proposing Release, supra note 11, at 80283.

³²⁹ See Proposing Release, supra note 11, at 80268-69.

³³⁰ See Proposing Release, supra note 11, at 80283.

exchanges, OTC market makers and ATSS, particularly as it relates to retail order flow;³³¹ and that it was “reasonable to assume that ... [applying] a minimum pricing increment to trades ..., could result in greater competition between exchanges and ATSS with other OTC market makers, including wholesalers...”³³² However, the Commission also stated that it could not anticipate how OTC market makers would adjust to increased competitive pressure and whether a market-wide trading increment would yield a “positive, negative or neutral” net effect on retail price improvement.³³³

The Commission received several comments on the proposal to establish minimum pricing increments for trades, including comments related to Quote and Trade Harmonization,³³⁴ Venue Harmonization,³³⁵ statutory authority,³³⁶ rationale,³³⁷ impact on price improvement,³³⁸

³³¹ See Proposing Release, supra note 11, at 80283.

³³² See Proposing Release, supra note 11, at 80303.

³³³ See Proposing Release, supra note 11, at 80326.

³³⁴ See, e.g., Nasdaq Letter I at 17; Citadel Letter I at 31; State Street Letter at 4; Citigroup Letter at 5; MFA Letter at 7, Letter from Nandini Sukumar, Chief Executive Officer, World Federation of Exchanges, dated Mar. 30, 2023 (“World Federation of Exchanges Letter”) at 5, STA Letter at 7, NYSE Letter I at 6; Nasdaq Letter I at 17, NYSE, Schwab, and Citadel Letter at 2, Brandes Letter at 2; Schwab Letter II at 6, Cambridge Letter at 6; B. Riley Letter at 1, Vanguard Letter at 5, Robinhood Letter at 40, 55; JPMorgan Letter at 6.

³³⁵ See, e.g., Cboe Letter II at 9, IEX Letter I at 2, Nasdaq Letter I at 2, Citigroup Letter at 5, Pragma Letter at 1, Luke Peterson Letter; Chris Miller Letter, Amanda Kappes Letter.

³³⁶ See Citadel Letter I at 4 and Robinhood Letter at 28.

³³⁷ See Citigroup Letter at 5, Nasdaq Letter I at 16, World Federation of Exchanges Letter at 4, Better Markets Letter I at 13, Vanguard Letter at 4, IEX Letter III at 5, Drew Ferguson Letter at 1, Max Garrison Letter at 1, Lukas Boller Letter at 1, Trent Miller Letter at 1, Phillip Worts Letter at 1, James Letter at 1, Andrew Garley Letter at 1, Larry Douglas Letter at 1, Steve Sullivan Letter at 1, Luke Czarnota Letter at 1, Charles S Letter at 1, Melisa Virginillo Letter at 1, Melissa Hyer Letter at 1, Keagan Wethington Letter at 1, DH Letter at 1, Alex Riley Letter at 1, Trevor Capestany at 1, Marco Daebnitz at 1, Steven Sullivan Letter at 1. See also Patrick Sexton, EVP, General Counsel & Corporate Secretary, Cboe Global Markets, Inc., dated Aug. 23, 2023 (“Cboe Letter III”) at 9.

³³⁸ See, e.g., SIFMA Letter II at 4, Citigroup Letter at 5, TradeStation Letter at 6, CCMR Letter at 27, Virtu Letter II at 7, Fidelity Letter at 13, BlackRock Letter at 9, Robinhood Letter at 20, JPMorgan Letter at 6, Morgan Stanley Letter at 4, TastyTrade Letter at 20 and Nasdaq Letter I at 18.

exceptions to minimum pricing increment for trades,³³⁹ and exchange RLP programs.³⁴⁰ After considering comments and in light of changes from the proposal that the Commission is making to amended Rule 612, the Commission has decided, consistent with one of the Reasonable Alternatives set forth in the Proposing Release,³⁴¹ not to adopt a minimum pricing increment for trades. As described above, the Commission is amending Rule 612 to adopt one smaller minimum pricing increment for quotes and orders that primarily focuses on those NMS stocks that are experiencing constraint with the \$0.01 minimum pricing increment. A secondary impact of the changes to the minimum quoting increment should be that it helps to partially address the concerns related to fair competition between exchanges, ATSS, and OTC markets that proposing a market-wide trading increment was designed to address.

Amended Rule 612 will reduce the minimum pricing increment for quotes and orders to \$0.005 for certain NMS stocks that are priced equal to, or greater than, \$1.00 per share which in turn also effectively reduces the increment that such stocks are able to trade in. Under amended Rule 612, OTC markets will continue to be able to trade more readily in comparatively smaller increments (e.g., \$0.001 or \$0.0001) than exchanges and ATSS, however, exchanges and ATSS will now be able to trade more regularly at smaller increments (i.e., \$0.005 or \$0.0025), compared to preexisting Rule 612, for those NMS stocks that are assigned the \$0.005 minimum pricing increment. By effectively reducing the trading increment for such NMS stocks, which represent a significant amount of the daily trading volume (approximately 58%) and dollar

³³⁹ See, e.g., JPMorgan Letter at 5; ICI Letter I at 17-18, Vanguard Letter at 5; IEX Letter I at 17 and BlackRock Letter at 9.

³⁴⁰ See, e.g., Cboe Letter III at 10-11, IEX Letter I at 17-18, JPMorgan Letter at 6, Ontario Teachers et al. Letter at 2, NYSE Letter I at 6.

³⁴¹ See Proposing Release, supra note 11, at 80339.

volume (approximately 43%),³⁴² the potential trade pricing discrepancy between exchanges, ATSS and OTC markets, while not harmonized, will be reduced. Market participants will be able to better compete based on the pricing of quotes and orders. Thus, because reducing the minimum quoting increment for a significant amount of volume of NMS stocks, whether measured by trading or dollars, also effectively reduces the trading increments for those stocks, the concerns related to the ability of OTC market makers to trade in comparatively finer increments raised by the Commission in the Proposing Release will be partially addressed so that the Commission has determined not to adopt the minimum pricing increment for trading.

However, because the amendment to Rule 612 only partially addresses the competitive dynamic between OTC market makers and exchanges and ATSS described in the Proposing Release, and furthermore allows OTC market makers to continue to execute trades in comparatively finer increments, the Commission staff will continue to monitor sub-penny trading to evaluate whether further action is appropriate for the protection of investors and to assure “fair competition among brokers and dealers, among exchange markets, and between exchange markets and markets other than exchange markets” in the national market system.

The Commission’s simplified, incremental approach to amending Rule 612 focuses on addressing issues that have developed regarding quoting constraints for certain NMS stocks because of the \$0.01 minimum pricing increment. Further, the amendment to Rule 612 will facilitate the transition of market participants and investors to the new tick size regime and the wider use of sub-penny quoting. The amendment, as adopted, also reduces the anticipated implementation costs compared to the proposed amendments to Rule 612. Finally, under

³⁴² See infra section VII.D.1.a.

amended Rule 612: (1) RLPs³⁴³ that operate pursuant to Commission exemptions that either permit certain quoting and trading in increments of \$0.001,³⁴⁴ or aggregate order flow at the midpoint,³⁴⁵ will be able to continue to operate without interruption and without changes to exchange rules or the grant of further exemptive relief by the Commission; (2) sub-penny price improvement will continue to be permitted consistent with the requirements of the rule; (3) and investors will continue to be able to manage their order flow and implement trading strategies through the use of midpoint orders and benchmark trades.

IV. Final Rule 610 of Regulation NMS - Fees for Access to Quotations

The Commission is adopting amendments to Rule 610(c)(1)(ii) as proposed with technical modifications to remove the reference to the minimum pricing increment. The Commission is not adopting proposed Rule 610(c)(1)(i) because it is unnecessary.³⁴⁶ Further, the Commission is adopting amendments to Rule 610(c)(2) as proposed with modifications to align the access fee caps for protected quotations in NMS stocks priced below \$1.00 and those priced \$1.00 and above.³⁴⁷ Finally, the Commission is removing outdated references to the “The Nasdaq Stock Market, Inc.” in Rule 610(c), as proposed, because the Nasdaq Stock Market is now a national securities exchange and the language is redundant.³⁴⁸

³⁴³ See supra note 146.

³⁴⁴ See, e.g., NYSE Rule 7.44 and BX Rule 4780.

³⁴⁵ See, e.g., NYSE Arca Rule 7.44-E and IEX Rule 11.232.

³⁴⁶ The Commission is not adopting the proposed 5 mils access fee cap because it is not adopting the \$0.001 minimum pricing increment. As proposed, all protected quotes in NMS stocks priced \$1.00 or more that would have been assigned a minimum pricing increment other than \$0.001 would have been subject to the proposed 10 mils access fee cap. The Commission is adopting this same model – all protected quotes in NMS stocks priced \$1.00 or more will be subject to the 10 mils access fee cap.

³⁴⁷ See infra section VII.D.2.a.

³⁴⁸ See Proposing Release, supra note 11, at 80292.

Specifically, under Rule 610(c), as amended, a trading center³⁴⁹ will not be permitted to impose, or permit to be imposed, any fee or fees for the execution of an order against a protected quotation of the trading center or against any other quotation of the trading center that is the best bid or best offer of a national securities exchange or the best bid or best offer of a national securities association in an NMS stock that exceed or accumulate to more than \$0.001 per share if the price of the protected quotation or other quotation is \$1.00 or more, and the fee or fees will not be permitted to exceed or accumulate to more than 0.1% of the quotation price per share if the price of the protected quotation or other quotation is less than \$1.00.

The Commission is also adopting Rule 610(d) as proposed to require that all exchange fees and rebates be determinable at the time of execution. For the reasons discussed below, these amendments to Rule 610 are appropriate for the modern national market system.

A. Background

Rule 610(c) was adopted in furtherance of the Congressional directives in section 11A of the Exchange Act and was designed to promote fair and non-discriminatory access to quotations displayed in the national market system.³⁵⁰ Rule 610(c) seeks to ensure the fairness and accuracy of displayed quotations by establishing an outer limit on the cost of accessing such quotations³⁵¹ and was designed to help to ensure that orders placed in the national market system reflect the best prices available. The access fee caps are necessary to achieve the purposes of the Exchange Act, including section 11A(c)(1)(B) of the Exchange Act, which authorizes the Commission to

³⁴⁹ 17 CFR 242.600(b)(106).

³⁵⁰ The Commission also stated that by imposing a uniform fee limitation of \$ 0.003 per share, Rule 610(c) will promote equal regulation of different types of trading centers. See Regulation NMS Adopting Release, supra note 4, at 37595.

³⁵¹ See Regulation NMS Adopting Release, supra note 4, at 37502.

adopt rules assuring the fairness and usefulness of quotation information. The Commission has stated that for quotations to be fair and useful, “there must be some limit on the extent to which the true price for those who access quotations can vary from the displayed price.”³⁵²

Rule 610 was adopted at the same time as Rule 611, the Order Protection Rule, which established intermarket protection against trade-throughs³⁵³ for all NMS stocks. Rule 610(c) was designed to preclude trading centers that posted protected quotations from raising their fees in an attempt to take improper advantage of the trade-through protections adopted under Rule 611.³⁵⁴ The Commission designed the access fee caps to preserve the benefits of the strengthened price protection under Rule 611 and more efficient linkages among trading centers that were developed under Regulation NMS to access protected quotations that could be disrupted if substantial fees were charged.³⁵⁵ At the time of adoption, the Commission recognized the importance of protecting the best displayed and accessible prices in promoting deep and stable markets that minimize investor costs. In this regard, the Commission stated that Rule 611 would help to minimize investor transaction costs, which is “the hallmark of efficient markets” and a “primary objective of the [national market system].”³⁵⁶ Rule 610 is an important component in supporting these goals.

Market participants, including investors, need fair and efficient access to the best priced quotations in the national market system. Therefore, Rule 610(c) remains an important part of the

³⁵² See Regulation NMS Adopting Release, supra note 4, at 37545.

³⁵³ A trade-through occurs when a trading center executes an order at a price that is inferior to the price of a protected quotation that is displayed by another trading center. See 17 CFR 242.600(b)(105) for the definition of trade-through under Regulation NMS.

³⁵⁴ See Regulation NMS Adopting Release, supra note 4, at 37544 and 37595.

³⁵⁵ See Regulation NMS Adopting Release, supra note 4, at 37545.

³⁵⁶ See Regulation NMS Adopting Release, supra note 4, at 37498.

national market system to achieve the purposes of the Exchange Act, preserve the benefits of price protection, and help ensure that displayed quotations reflect something close to actual costs incurred for the transaction.³⁵⁷

B. Issues Raised in the Existing Market Structure and the Need for the Amendments

The national market system of 2024 is significantly different than the national market system that existed when Regulation NMS was adopted in 2005.³⁵⁸ Since Regulation NMS was adopted, new trading practices, order types, and routing strategies have developed that did not exist when Rule 610 was adopted.³⁵⁹ In addition, exchanges have developed complex fee structures that charge the outer limits permitted for accessing protected quotations and use those fees to fund rebates, which have the effect of creating a discrepancy between displayed prices and net prices.³⁶⁰ Finally, the national market system has seen a proliferation of new exchanges, often within the same exchange group, that implement varied pricing models to attract specific

³⁵⁷ The access fee caps were calculated based upon the then current fees that were charged by certain trading venues. See Regulation NMS Adopting Release, supra note 4, at 37545 (stating “the \$0.003 fee limitation is consistent with current business practices, as very few trading centers currently charge fees that exceed this amount . . . [and those that do] do not account for a large percentage of the trading volume.”). At the time the access fee caps were adopted, the minimum pricing increment for quotes and orders priced \$1.00 or greater was \$0.01 as Rule 612 was adopted at the same time as Rule 610(c).

³⁵⁸ See Proposing Release, supra note 11, at 80289. See also Letter from John Ramsay, Chief Market Policy Officer, Investors Exchange LLC, dated Oct. 19, 2023 (“IEX Letter IV”) at 5-10; Nasdaq Letter I at 4.

³⁵⁹ See also Letter from Theodore R. Lazo, Managing Director & Associate General Counsel, SIFMA, to Brent J. Fields, Secretary, Commission, dated Mar. 29, 2017, at 8 (stating exchanges have developed order types primarily designed to avoid paying high fees, market participants implement complex routing strategies (consistent with their best execution obligations) to avoid paying high exchange access fees in favor of lower costs ATs and the market place has seen a high level of fragmentation “driven by each exchange group’s desire to provide a variety of pricing models within the wide pricing range between 0 and 30 mils.”) (“SIFMA 2017 Letter”).

³⁶⁰ See infra section VII.C.2, note 1107 and accompanying text and note 1457 (showing that the primary reason that access fees remain near 30 mils on most exchanges is to fund rebates) and Panel A of Table 4, infra section VII.C.2.c.

market participants to their markets.³⁶¹ The Commission has monitored these developments and engaged extensively with market participants about the impact of the modern fee structures on fair and efficient access to protected quotations as well as the usefulness and accuracy of such quotations.³⁶²

Market participants have considered and suggested reductions in the access fee caps for many years to address market distortions (as further discussed below)³⁶³ and better reflect evolutions in the market since Regulation NMS was adopted.³⁶⁴ One commenter stated that

³⁶¹ See *infra* Tables 4 and 5 showing within the large exchange groups multiple exchanges each with different fee and rebate models. See also e.g., SIFMA 2017 Letter, *supra* note 359 at 8 (stating “the high level of fragmentation . . . is in part driven by each of the exchange group’s desire to provide a variety of pricing models within the wide pricing range between 0 and 30 mils.”).

³⁶² For example, the EMSAC considered, among other things, whether the access fee cap should be modified. See EMSAC Archives, *supra* note 4. See also Concept Release on Equity Market Structure, *supra* note 59.

³⁶³ See, e.g., *infra* notes 371-376 and accompanying text and *infra* section IV.B.2 and IV.D.

³⁶⁴ See, e.g., Letter from Theodore R. Lazo, Managing Director & Associate General Counsel, SIFMA, to Brent J. Fields, Secretary, Commission, dated May 24, 2018, at 2 (commenting on File No. S7-05-18 “Transaction Fee Pilot for NMS Stocks”) (stating “On several occasions, SIFMA has recommended that the Commission reduce the access fee cap to no more than five cents per 100 shares because the cap has not been adjusted to reflect market developments since Regulation NMS was adopted more than a decade ago.”) (“SIFMA 2018 Letter”); Goldman 2018 Letter at 1 (stating “a reduction in the Fee Cap from \$0.0030 to \$0.0010 per share could be supported today [2018] and would be better calibrated with the present-day trading and execution costs, which have decreased substantially since 2005”); SIFMA 2017 Letter, *supra* note 359, at 3 (stating the Commission should consider “reducing the access fee cap to no more than \$0.0005 for all securities” because “[s]ince Reg. NMS was adopted, spreads have narrowed and commissions have decreased, making the existing cap of access fees outsized relative to today’s market realities.”); Letter from Theodore R. Lazo, Managing Director & Associate General Counsel, SIFMA, to Mary Jo White, Chair, Commission, dated May 24, 2015, at 2-3 (supporting reduction in access fee to “no more than five cents per hundred shares” because access fees are “an outsized element of transaction costs that in turn distorts price discovery and contributes to market complexity, both on- and off-exchange” and further stating “market participants regularly implement complex order routing strategies . . . that divide, route and re-route orders and parts of orders, when possible, to market centers that enable them to avoid paying excessive access fees” and also stating “access fees have increased complexity on exchanges . . . through the proliferation of exchange order types designed to avoid access fees.”); Letter from Daniel Keegan, Managing Director, Head of Americas Equities, Citigroup Global Markets Inc. to Elizabeth Murphy, Secretary, Commission, dated Aug. 7, 2014, at 5 (“Citigroup 2014 Letter”) (commenting on Concept Release on Equity Market Structure and stating “[t]he SEC should explore the impacts on the overall markets of complex market structure issues such as maker/taker pricing and access fees” and “[a]t a minimum, the cap on access fees should be reduced to below 10 mils”); Letter from Theodore R. Lazo, Director & Associate General Counsel, SIFMA, to Mary Jo White, Chair, Commission, dated Oct. 24, 2014, at 2 (providing “Recommendations for Equity Market Structure Reforms” including “that exchange access fees be significantly reduced, to no more than five cents per 100 shares”); NYT Dealbook OpEd: